

Traffic Sign Detection Using YOLOv8

Impact of Data Augmentation on GTSDb Detection Performance

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2. Abstract

This report investigates the effect of image-based data augmentation on the performance of a YOLOv8s object detection model training for traffic sign recognition using the German Traffic Sign Detection Benchmark (GTSDb). Two experimental conditions were compared: a baseline model trained on the original dataset of approximately 500 high-quality road-scene images, and an augmented model trained on an expanded dataset produced using Roboflow's augmentation pipeline. The augmentation strategy simulated harsh driving conditions using blur (2px), random brightness variation (-20% to +20%), and additive noise (1.13x).

Both models used the YOLOv8s architecture (73 fused layers, 11.1 M parameters, 28.5 GFLOPs) and were trained for 30 epochs at 1024x1024 px resolution. Evaluation was conducted on a test set of 89 images containing 148 annotated instances across 43 sign classes. The augmented model achieved mAP50 of 0.65 and mAP50-95 of 0.560 versus the baseline's 0.610 and 0.519, respectively, with a particularly significant improvement in recall (0.658 vs 0.501). These results support the hypothesis that augmentation improves generalisation in constrained, small-object detection datasets.

3. Introduction

Problem Statement:

Reliable, real-time traffic sign detection is a foundational capability for autonomous vehicles and self-driving cars. Signs communicate legally binding instructions that, if broken, can lead to direct safety consequences for both the passengers and pedestrians. The problem is technically challenging: signs are very small relative to an image field of view, appearance varies with the viewing angle and distance, and real-world imaging introduces blur, glare, and sensor noise.

Why The Problem Matters:

Traffic sign detection is important for safety-critical systems to be deployed at scale. Beyond autonomous driving, accurate sign detection supports infrastructure monitoring, mapping, and compliance checking. Training robust detectors under limited data budgets is therefore of broad practical importance.

Overview of Approach:

This study uses the YOLOv8s model that is trained under two conditions: a baseline using original GTSDb images, and an augmented variant using Roboflow, thereby expanding the

training data. Images were resized to 1024x1024 px because prior tests showed extremely poor performance at evaluating showed signs at 640 px; there were just too few pixels for the model to learn any features.

4. Related Work and Background

YOLO Architecture:

The YOLO family processes entire images in a single forward pass, predicting bounding boxes and class probabilities simultaneously. YOLOv8 introduces an anchor-free detection head, CSPDarknet backbone, and improved PANet neck, meaning it is much better at small-object detection, which is relevant for small traffic signs.

The GTSDDB Dataset:

The German Traffic Sign Detection Benchmark (Houben et al., 2013) consists of 600 full road scene images at 1360x800 px, approximately 500 of which contain annotated signs. The signs span 43 classes that classify what the sign is in detail. The GT file in the dataset also draws bounding boxes on each of the signs.

Data Augmentation for Object Detection:

Augmentation has become standard practice for training on limited datasets. For detection, augmentations must be applied consistently to both image and bounding box coordinates. Augmentations like brightness, noise, and blur improve robustness to sensor and environmental variation.

5. Methodology

Dataset Description:

Raw GTSDDB PPM images were converted to JPEG, and annotations in the gt.txt file were converted from absolute pixel coordinates (left, top; right; bottom) to YOLO normalised format. Images with no gt.txt entries were strictly excluded from the training. After cleaning, 443 annotated images were retained and split into train/validation//test subsets via Roboflow.

Split	Images	Instances
Train	312	~983
Validation	89	~280
Test	42	~132

Table 1: Dataset split summary

Augmentation Strategy:

Augmentation was chosen to simulate harsh, degraded real-world imaging conditions:

- Blur: 2px - models lens defocus and motion blur at different speeds
- Brightness Variation: $\pm 20\%$ - covers different lighting conditions, like nighttime, morning, overcast, etc.
- Noise: 1.13x - approximates sensor noise in low lighting environments

Spatial augmentations (flips, rotations) were avoided as traffic signs are orientation-specific. All augmentations were applied only to training images; the validation and test sets were unmodified.

Model Architecture and Training:

YOLOv8s (73 layers, 11,141,451 parameters, 28.5 GFLOPs) was trained under identical hyperparameters for both conditions: 30 epochs, 1024x1024 px input. The only difference between conditions was the training dataset.

Input Resolution Rationale:

At 640px, many GTSDb signs show up as 4-8 pixels after downsampling, which is far below the effective receptive field of the detection head. Upscaling the images to 1040px preserves the spatial detail for feature extraction, at the cost of about 3 increases FLOPs. This tradeoff was necessary as the model was not performing well and was not learning any features with the 640x640px images.

6. Experiments and Results

Overall Performance

Table 2 presents headline evaluation metrics for both models on the test set. The augmented model outperforms the baseline across all four primary metrics.

Metric	Baseline	Augmented	Δ Absolute	Δ Relative
Precision (P)	0.687	0.570	-0.117	-17.0%
Recall (R)	0.501	0.658	+0.157	+31.3%
mAP50	0.610	0.650	+0.040	+6.6%
mAP50-95	0.519	0.560	+0.041	+7.9%
Inference (ms)	12.5	12.1	-0.4	-3.2%

Table 2: Overall performance comparison on the GTSDb test set(42 images).

The augmented model's lower precision alongside substantially higher recall reflects a shift in operating point: the model produces more true positives at the cost of a moderate increase in false positives. In a safety-critical detection context, this trade-off is generally favourable since missed detections carry a higher cost than false alarms.

Precision-Recall and Confidence Curves

Figures 1-3 show the Precision-Recall, Precision-Confidence, and F1-Confidence curves for the baseline model. The PR curve confirms an $mAP@0.5$ of 0.610, with precision starting high (~ 0.73) and degrading as recall extends towards 1.0. The F1-confidence curve peaks at 0.43 at a confidence threshold of 0.347, indicating that it is the optimal operating point for the baseline model.

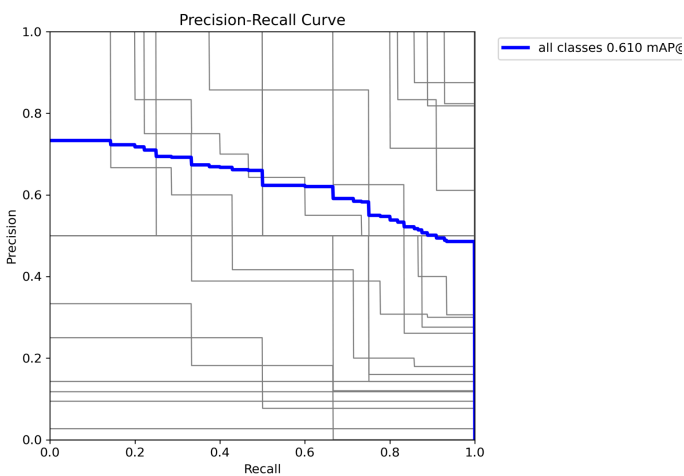


Figure 1

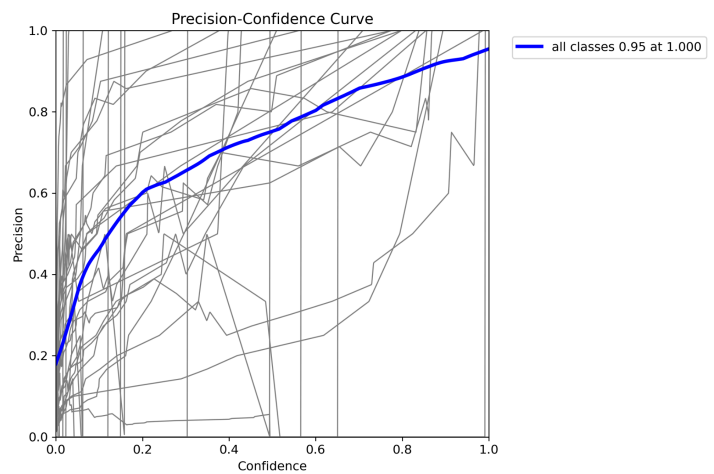


Figure 2

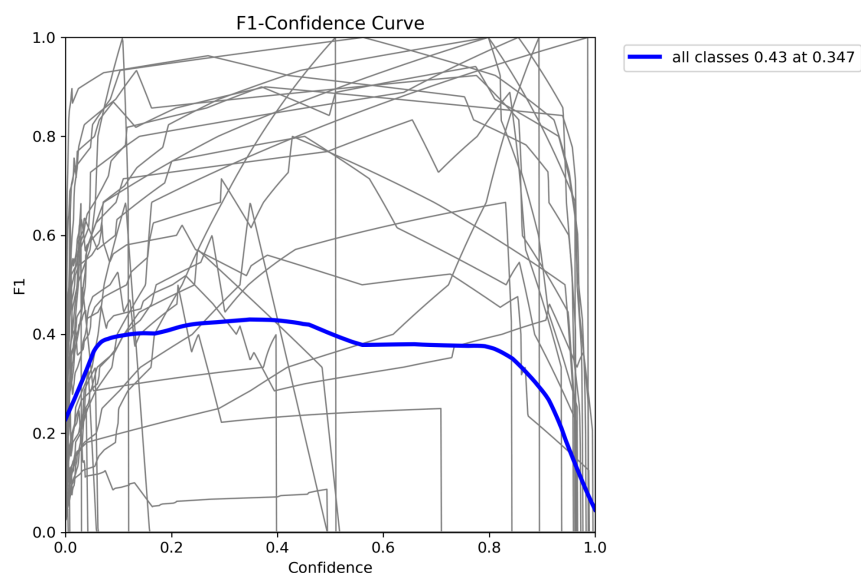


Figure 3 ->

Figure 1 (Baseline): Precision-Recall curve - $mAP@0.5 = 0.610$. The curve starts at precision ~ 0.73 and maintains relatively high precision up to recall ~ 0.5 before degrading

Figure 2 (Baseline): Precision-confidence curve - maximum precision 0.95 at confidence threshold 1.0. Precision rises steeply and consistently with increasing confidence threshold.

Figure 3 (Baseline): F1-Confidence curve, the peak F1 = 0.43 at confidence threshold 0.347. The flat portion between 0.1 and 0.5 suggests the model is relatively stable across a wide confidence range.

Figures 4-6 show the equivalent curves for the augmented model. The PR curve confirms $mAP@0.5$ of 0.650, with precision starting higher (~ 0.78) and maintaining above 0.65 until recall ~ 0.5 . The F1 Curve peaks at 0.49 at threshold 0.093, which is higher than the baseline's 0.43, confirming the augmented model achieves better F1 balance. The precision curve reaches 0.97 at full confidence, higher than the baseline's 0.95.

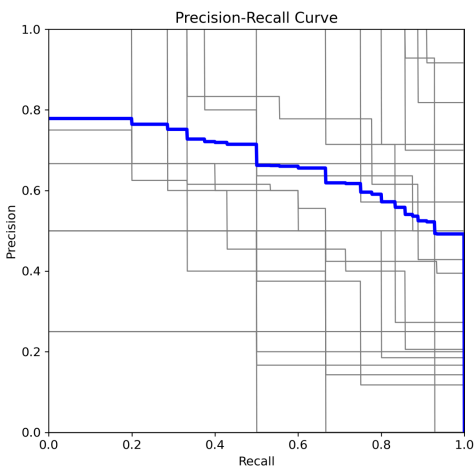


Figure 4

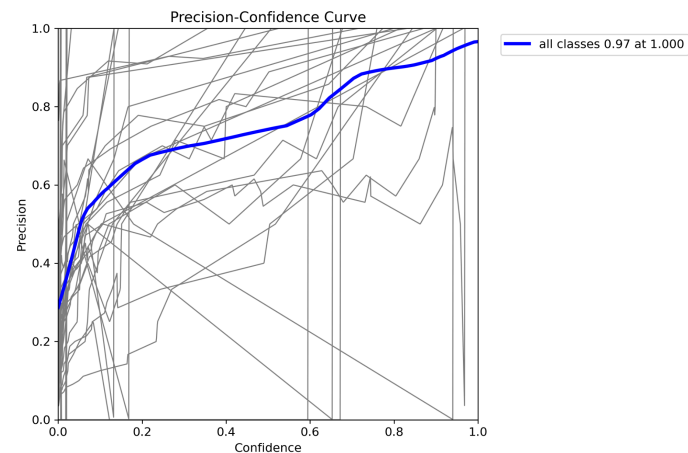


Figure 5

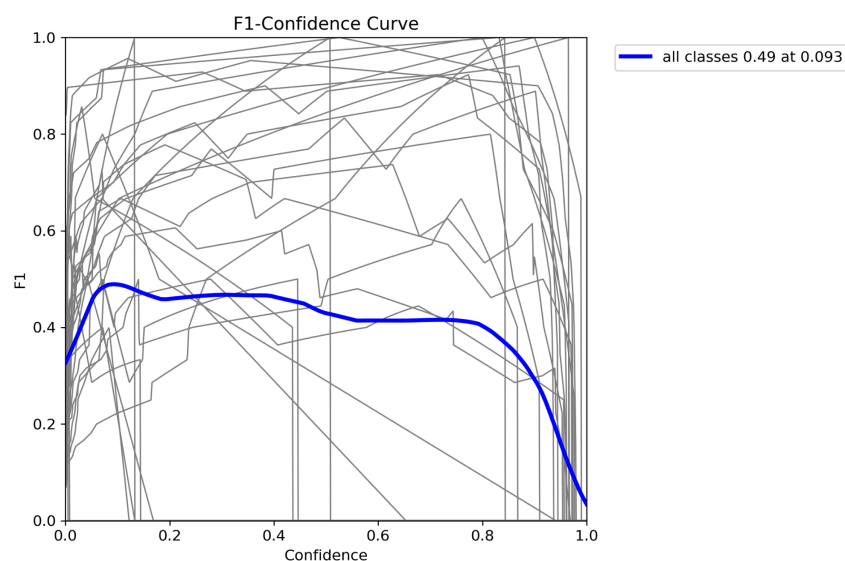


Figure 6 ->

Figure 4 (Augmented): Precision-Recall curve - $\text{mAP}@0.5 = 0.650$. The augmented model maintains higher precision across the low-recall range compared to the baseline, indicating more confident true positive detections.

Figure 5 (Augmented): Precision confidence curve shows a maximum precision of 0.97 at confidence 1.0. The curve is steeper and starts higher than the baseline, reflecting the model's improved discrimination.

Figure 6 (Augmented): F1-Confidence curve, which peaked at 0.49 at a confidence threshold of 0.093. The augmented model achieves a 14% relative improvement in peak F1 over the baseline (0.43).

Training Loss Curves:

Figures 7 and 8 show the full 30-epoch training curves for both models, including training/validation losses and metric trajectories. Tables 3 and 4 provide the sampled numerical values at 5-epoch intervals.

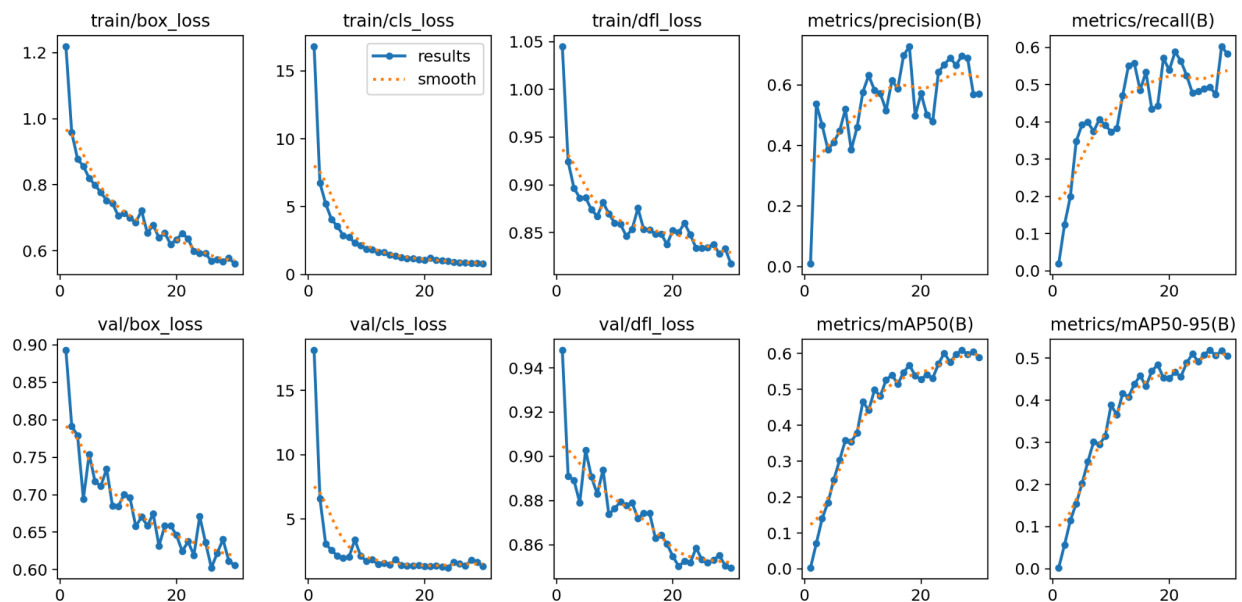


Figure 7 (Baseline): Training curves across 30 epochs. All three loss components decline as time goes on. Classification loss drops sharply from 16.8 at epoch 1. $\text{mAP}50$ rises steadily, peaking around 0.59-0.61 after epoch 23.

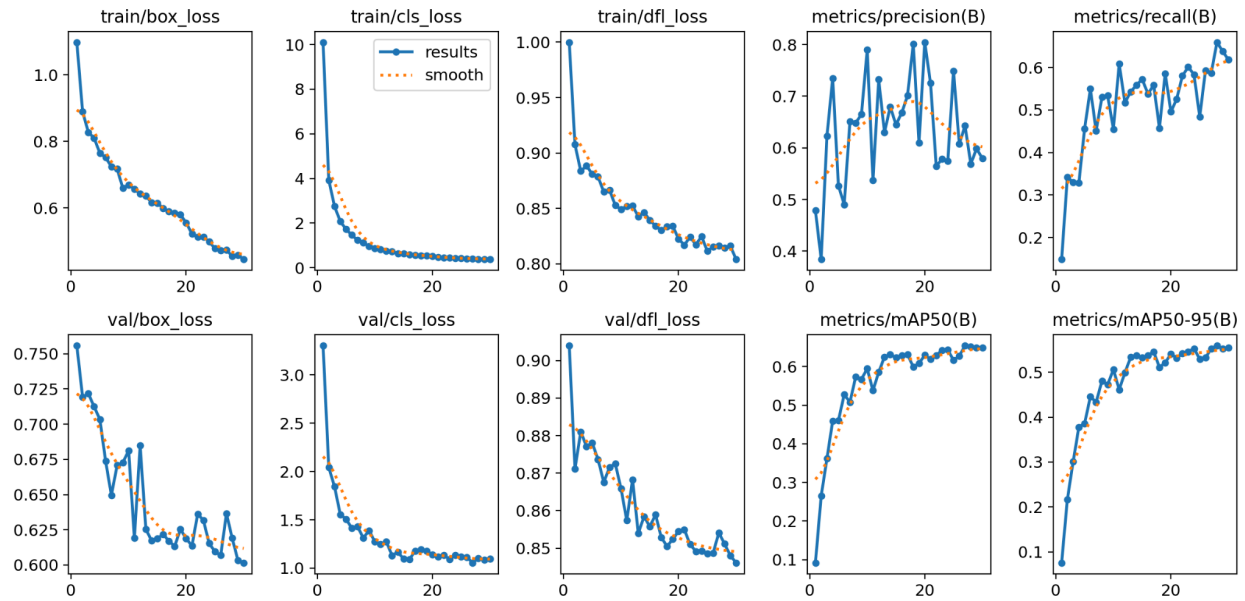


Figure 8 (Augmented): Training curves across 30 epochs. The augmented model’s classification loss starts lower (10.1 vs 16.8) and converges faster. mAP50 rises more smoothly, reaching 0.649 by epoch 30 compared to the baseline’s 0.589.

Confusion Matrices

Figures 9 and 10 show the normalised confusion matrices for both models. The diagonal represents correct detections, and the off-diagonal cells indicate inter-class confusion.

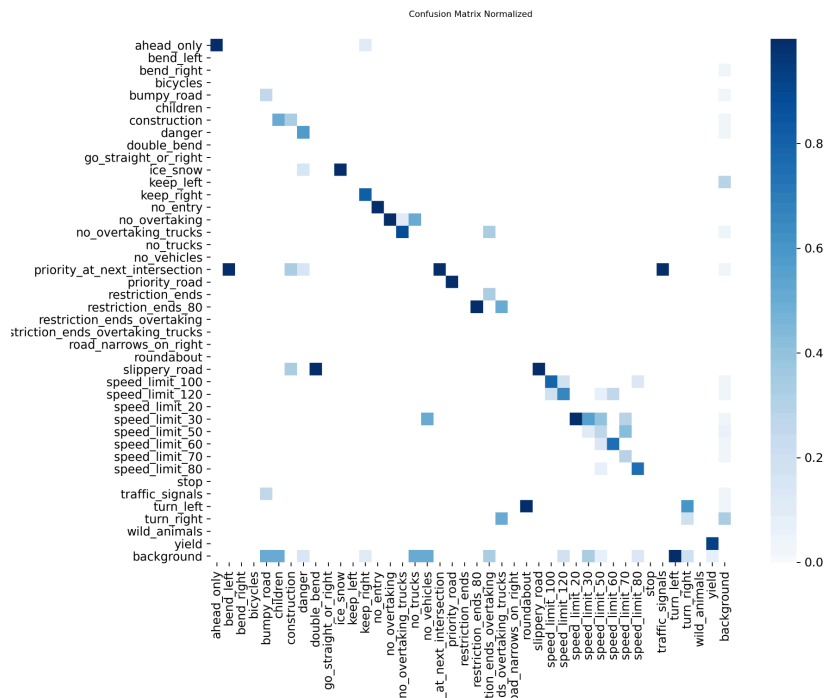


Figure 9 (Baseline): Confusion matrix across 43 classes. As we can see, some classes are severely underrepresented while others are more frequent, which makes sense in a real-world scenario.



Figure 10 (Augmented): confusion matrix for the augmented model. The diagonal is denser for speed limit sub-classes compared to the baseline, meaning it is better at differentiating between the similarly shaped speed limit sign family.

7. Discussion

What Worked Well:

The augmentation strategy produced consistent gains in recall (+31.3%) and mAP at both IoU thresholds. The intuition, therefore, is clear: training on degraded examples forces the model to learn structural and geometric sign features rather than relying on brightness or contrast patterns that may not persist in deployment.

The augmented model's classification loss converges from a lower starting point (10.1 vs 16.8 at epoch 1), suggesting that richer training examples provide for a more consistent gradient signal for the classification branch. The F1-curve improvement (0.43 -> 0.49 peak) and higher precision-at-confidence of 0.97 vs 0.95 confirm the augmented model makes more reliable predictions when it does commit to a detection.

The choice of 1024 px input resolution was validated empirically. Many GTSDb signs occupy 30-80 px at native resolution, downsampling them to 640px would have reduced this to an even smaller pixel size, which would be unrecognizable for the model.

Limitations and Challenges

Class imbalance: Test coverage ranges from 1 to 15 instances per class. Single instance test classes produce uninformative binary metrics. Stratified sampling or a larger dataset would enable more reliable conclusions.

No spatial augmentation: Geometric augmentations (crop and scale, perspective distortion) could further improve robustness to viewing angle variation without compromising sign orientation semantics.

No threshold calibration: The precision-recall trade-off was not explicitly optimised. Lowering the confidence threshold on the augmented model from the default towards 0.093 (peak F1) could further improve recall without significant precision loss.

Key Insights

The biggest insight is that augmentation shifts the precision-recall operating point. The model becomes more willing to detect (higher recall) at the cost of some false positives (lower precision). For traffic sign detection, missed stop signs carry a much higher cost than false alarms.

The confusion matrices reveal that inter-class confusion within the speed limit family (30,50,60,70,80,100,120) remains the dominant reason for error, particularly for the baseline. The augmented model shows tighter diagonal concentration in this cluster, which is a benefit of learning under varied environmental conditions.

8. Conclusion and Future Work

Conclusion:

This study demonstrated that a targeted augmentation pipeline (blur, brightness variation, and additive noise) improves the YOLOv8s model's detection performance on the GTSDb dataset. The augmented model achieved mAP50 of 0.650 vs. 0.610 for the baseline, with particularly strong recall improvement (+31.3%) and a higher peak F1 score (0.49 vs 0.43). These gains were obtained without architecture changes, hyperparameter tuning, or additional real-world data, representing a cost-effective robustness strategy. Furthermore, the results highlight the importance of input resolution selection for small-object detection and the challenge of rare class coverage under dataset constraints.

Future Work:

Superclass model: Training on the three GTSDb superclasses (prohibitive, danger, mandatory) reduces imbalance and may improve detection of unrepresented fine-grained classes.

Larger backbone: Upgrading from the YOLOv8s to YOLOv8m would add more capacity for fine-grained sign classification.

9. References

Houben, S., Stallkamp, J., Salmen, J., Schlipsing, M., & Igel, C. (2013). Detection of traffic signs in real-world images: The German Traffic Sign Detection Benchmark. IJCNN, 1–8.

Jocher, G., Chaurasia, A., & Qiu, J. (2023). Ultralytics YOLOv8.
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